Finite normal pairs as incidence geometries

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Abstract

We give a complete reconstruction of the finite normal-pair representation results of Díaz Ramos, Molinier and Viruel. The common paper supplies the complete ordered-pair graph encoding of a finite group and a normal subgroup, and its chamber theorem turns that graph into a connected incidence geometry. We apply those full shared proofs with their exact rank, count and connectedness statements. We also recover the authors' smaller recursive (S_n, A_n) family, including its color sizes and faithful top action. A completed-hexagon example is calculated directly. Its corrected type-preserving subgroup, the chamber degree correction and the degree-one base of the recursive family are kept separate from the unchanged principal representation theorem.

1 Finite normal pairs from the common construction

The complete ordered-pair encoder and its recognition, rigidity, extension, color-kernel and chamber-completion proofs now belong to the common paper [2]. They are the constructions of Díaz Ramos, Molinier and Viruel [1, Theorem 3.1 and Theorem A], with all hypotheses and exceptional parameters retained. The present application records both full source statements before treating the additional recursive family and completed hexagon.

Theorem 1. *Every finite normal pair $H \trianglelefteq G$ is isomorphic to $(C(\Gamma), K(\Gamma))$ for a finite connected incidence geometry Γ. For $G \neq 1$ the construction below has rank $[G : H] + 3$; for $G = 1$ it has rank one. No fixed rank-two assertion is made.*

Proof. Apply common Theorem UR.FinitePairs to the finite pair (G, H). It includes the explicit one-object geometry for $G = 1$; otherwise its ordered-pair graph is connected, triangle-free and has minimum degree two, so the common chamber theorem gives the stated connected geometry and rank. □

Theorem 2 (Ordered-pair encoding)**.** *Let $G \neq 1$ be finite and $H \trianglelefteq G$. There is a connected properly colored triangle-free graph $X(G, H)$ of minimum degree two with*

$$(C(X(G,H)), K(X(G,H))) \cong (G, H).$$

For $q = |G|$ it has $(q^3 + 9q^2 - 8q)/2$ vertices and $[G : H] + 3$ colors.

Proof. Apply common Theorem UR.PairEncoding with $q = |G| \geq 2$. Its full ordered edge list, intrinsic degree recognition and unique extension of right translations prove the exact pair, graph hypotheses, vertex count and color count, including $q = 2$. □

Normality is necessary because the type-preserving group is the kernel of the palette action. In particular G/H must embed in the symmetric group on that palette; rank two cannot realize arbitrary indices. The complete arbitrary-input cleaning/conversion theorem is also in the common paper. It preserves components and does not silently make a disconnected input connected.

2 The recursive symmetric-group family

The smaller family in [1, Example 5.2] contains explicit information not implied by the bare existence theorem. We retain its construction, color counts and faithful top action.

Theorem 3. *For each $n \geq 2$ there is a connected properly colored triangle-free graph X_n, with colors $0, \dots, n$, satisfying*

$$(C(X_n), K(X_n)) \cong (S_n, A_n).$$

Its color sizes are

$$|t^{-1}(n)| = n, \qquad |t^{-1}(i)| = \frac{n!}{2(i-1)!} \quad (0 \leq i < n), \tag{1}$$

where $(-1)! = 0! = 1$. The top class carries the faithful natural S_n-action. An odd permutation exchanges exactly colors zero and one, while an even permutation fixes every color. The minimum degree is at least two for $n \geq 3$.

Proof. Let X_2 be the four-vertex path with successive colors $2, 0, 1, 2$. Label its endpoints $1, 2$. Its only correlations are the identity and the reversal; reversal exchanges colors zero and one. This gives the claimed pair and faithful top action. The endpoints have degree one, so this is a pre-induction object, not a minimum-degree-two base case.

Suppose X_{n-1} has been constructed, with its top vertices labeled $1, \dots, n-1$. Introduce vertices $v_1, \dots, v_n$ of color n and vertices $v_{\{i,j\}}$ of color $n-1$, joined to v_i and v_j. Choose even permutations $g_j \in A_n$ with $g_j(n) = j$ and $g_n = 1$; these exist because A_n is transitive for $n \geq 3$. At each j attach a copy of X_{n-1}, identifying its top vertex labeled i with $v_{\{g_j(i),j\}}$. The interiors of different copies are disjoint, and all their lower colors are inherited.

The graph is connected: each copy is connected and meets its boundary, and the new top and pair vertices form the subdivided complete graph. Within every copy the coloring is proper and there is no triangle. Different copies meet only at pair vertices, all of the same color and hence pairwise nonadjacent. The neighbors of a new top vertex also all have color $n-1$ and are pairwise nonadjacent. Thus no triangle is introduced. An interior vertex retains its old degree. A top vertex has degree $n-1$; a pair vertex gets its two new top neighbors and its neighbors in two copies. For $n = 3$ only pair vertices had been endpoints of the base paths: their new degree is four, whereas the other vertices have degree two. For larger n the minimum-degree assertion follows by induction.

There are n top vertices and $\binom{n}{2}$ pair vertices. For $i < n-1$, the class of color i is the disjoint union of n copies of the previous class. These recurrences and the base path prove (1).

For $\sigma \in S_n$ set $v_i \mapsto v_{\sigma(i)}$ and $v_{\{i,j\}} \mapsto v_{\{\sigma(i),\sigma(j)\}}$. On the copy at j use the inductively constructed correlation corresponding to

$$h_{\sigma,j} = g_{\sigma(j)}^{-1} \sigma g_j.$$

This permutation fixes n and therefore acts on the $n-1$ labels of that copy. Its sign equals the sign of σ, because the chosen representatives are even. The induction hypothesis says that all these local maps have the same palette permutation: the identity for even σ, or exchange of zero and one for odd σ.

On a boundary vertex labeled i in the copy at j, the image has global label

$$\{g_{\sigma(j)} h_{\sigma,j}(i), \sigma(j)\} = \{\sigma g_j(i), \sigma(j)\}.$$

Thus local maps agree with the displayed map on every shared pair vertex. They combine into a correlation, with inverse given by σ^{-1}. Moreover

$$h_{\tau,\sigma(j)} h_{\sigma,j} = h_{\tau\sigma,j},$$

so their composition respects multiplication. We have constructed an S_n-action with the stated palette kernel and natural action on the top.

It remains to exclude extra correlations. For $n \geq 4$, the top class of size n is uniquely smallest, since the next size is $\binom{n}{2} > n$ and all lower classes are larger. For $n = 3$ all four classes have size three, but the top vertices are precisely the degree-two vertices both of whose neighbors have degree four. Each other degree-two vertex has one neighbor of degree two and one of degree four. Therefore every correlation preserves the top class, in every case.

A correlation fixing the top pointwise fixes each pair vertex, since its two top neighbors characterize it. We verify the component-recognition hypothesis needed before applying induction to a copy. The graph X_m with its top removed is connected for every $m \geq 2$. For $m = 2$ it is an edge. For $m \geq 3$, it is the union of its m connected attached copies, and any two copies meet in their common pair vertex. Hence it is connected.

In X_n, deleting both top and pair vertices consequently leaves exactly the connected interiors of the attached copies. The boundary of the interior at j is

$$\{v_{\{i,j\}} : i \neq j\}.$$

Every boundary vertex meets the interior: in the base it is an endpoint of a path, and at every subsequent stage each top vertex has neighbors in the lower part. These boundary stars are distinct for $n \geq 3$. A correlation fixing all pair vertices must therefore preserve each copy. On that copy it is a correlation fixing the top pointwise, and is the identity by inductive faithfulness. This is the component-boundary rigidity argument of the common paper.

Thus restriction embeds the full correlation group in S_n. The constructed action already realizes every member of S_n, and its palette kernel is A_n. This proves all claims. □

For $n \geq 3$, the common chamber theorem immediately produces (S_n, A_n)-geometries of rank $n+1$. For $n = 2$, first clean the base path and then complete it, or use Theorem 1. The source's blanket minimum-degree claim must be qualified at this base: it is not legitimate to apply the chamber theorem directly to its degree-one endpoints.

3 A completion that enlarges the group

Number the vertices of a hexagon by $i \in \mathbb{Z}/6\mathbb{Z}$, color them by parity, and add a third-colored tip on each edge $\{i, i+1\}$ with i even. This is [1, Example 5.1]. The three tips have degree two and the six hexagon vertices degree three, so every correlation recognizes the hexagon. Its allowed dihedral symmetries are exactly the stabilizer of the alternating edge set. They are generated by $i \mapsto i+2$ and $i \mapsto 1-i$, and form S_3. The first generator preserves the two old colors and the second exchanges them; the pair is (S_3, C_3).

Now add a third-colored tip on all six edges. Every edge lies in its displayed triangle, and those triangles are the maximal cliques; the completed system is an incidence geometry. Old vertices now have degree four and tips degree two, so again the hexagon is intrinsic. Every dihedral symmetry extends uniquely to its incident tips and permutes the parity classes. Consequently the full correlation group is the dihedral group of order twelve. The color-preserving subgroup consists of rotations by even increments and reflections $i \mapsto 2a - i$. It has order six and is isomorphic to S_3. In particular

$$i \longmapsto -i, \qquad \text{tip on } \{i, i+1\} \longmapsto \text{tip on } \{-i-1, -i\} \tag{2}$$

is a nonidentity type-preserving involution.

Thus the completed pair is (D_{12}, S_3), with the subscript denoting group order. The source prints (D_{12}, C_3). Formula (2) alone already excludes the second entry C_3; the intrinsic hexagon argument computes the full pair. The intended obstruction survives: naive completion of this system acquires additional correlations. The initial system already has triangles, so it is outside the common chamber theorem. This example does not prove that each of that theorem's hypotheses is necessary in every possible alternative construction.

4 Source contract and certificates

The source is Díaz Ramos–Molinier–Viruel, arXiv:2511.02410v2, 16 December 2025. The full finite normal-pair theorem, the ordered-arrow construction, chamber completion, cleaning and the recursive symmetric family belong to those authors. Their proof acknowledges Frucht and König's classical graph-representation constructions. Those historical representation theorems are not unproved premises of the explicit constructions here.

The common paper and this application recover source Theorems 3.1, 4.1, 4.2 and A. They also recover the complete intended statements of Examples 5.1 and 5.2 with the stated corrections. All local recognition, extension, triangle-free, minimum-degree, connectedness and faithful-action arguments are supplied in the common paper and this application. The full ordered-pair proof and finite normal-pair existence theorem are shared in the common paper; neither application supplies an unpaid premise to the other. The three qualifications are the actual chamber degrees, the completed-hexagon type kernel, and the degree-one recursive base path. No optimality assertion about all completion hypotheses or all possible graph sizes is inferred. The family is asserted as (S_n, A_n); no universal identification of that pair with $(\mathrm{Aut}(A_n), \mathrm{Inn}(A_n))$ is needed or claimed. The source's Question 2.4 is answered by its own Theorem A; this recovery is not a newly solved conjecture.

The generic formal file for the common theorem proves the restriction and extension isomorphism for actual permutation subgroups and an embedded vertex set. It does not verify the gadgets that discharge its hypotheses. The unchanged `CompletedHexagonCounterexample.lean` gives four reported certificates: preservation of adjacency and color by (2), its order two, and the absence of a nontrivial involution in C_3. These are not a Lean computation of the full graph automorphism group or a formal proof for all finite normal pairs.

The two complete GAP programs in the appendix check the two hexagon groups, all 25 properly colored graphs on at most three labeled vertices under cleaning and completion, seven normal-pair examples, four completed pair examples, and the recursive family for $2 \le n \le 5$. They include exact restriction actions rather than comparing only group orders. They are bounded validation, since the general constructions and both exceptional group identifications have full written proofs in the common paper and this application. The full programs remain available; no diagnostic code is removed. Their drivers and proof infrastructure are recorded separately. There is no numerical compression claim in this edition.

References

[1] A. Díaz Ramos, R. Molinier and A. Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025.

[2] Human Compression Project, *Recognizable extensions of colored graphs*, accompanying common paper, Theorems UR.Restriction, UR.Chambers, UR.Cleaning, UR.PairEncoding and UR.FinitePairs.

[3] R. Frucht, *Herstellung von Graphen mit vorgegebener abstrakter Gruppe*, Compositio Mathematica **6** (1939), 239–250.

[4] D. König, *Theorie der endlichen und unendlichen Graphen*, Akademische Verlagsgesellschaft, Leipzig, 1936.